

CSCI-471
Assignment#1
Python Programming Assignment: Fitness Tracker Application

Objective:

Develop a Python-based Fitness Tracker Application that helps users keep track of their daily exercise routines, calories burned, and fitness goals. This project will involve using variables, data types, loops, conditionals, functions, and file handling.

Assignment Overview

You are required to build an application that allows users to:

1. Log exercises (type, duration, and calories burned).
2. View logged exercises for any given day.
3. Set fitness goals (e.g., calories to burn per week).
4. Track progress toward weekly goals.

Requirements:

1. Main Menu:
 - Create a text-based menu that offers the following options:
 1. Log an Exercise
 2. View Logged Exercises
 3. Set Weekly Calorie Goal
 4. Track Weekly Progress
 5. Exit
 - Use control flow (if-elif-else) to navigate the menu and perform the appropriate task.
2. Logging Exercises:
 - Users should be able to input:
 - Exercise type (e.g., running, cycling, yoga).
 - Duration (in minutes).
 - Calories burned (per minute).
 - Save this information to a file with the format:
 - Date | Exercise Type | Duration | Calories Burned.
 - Use lists to store daily exercises and file handling to save them persistently.
3. Viewing Logged Exercises:
 - Allow users to view their logged exercises for any given day by specifying the date.
 - Read the exercise log file and display all exercises for the specified date.

4. Setting a Weekly Goal:

- Users should be able to set a weekly calorie-burning goal (e.g., 3000 calories per week).
- Save this goal to a file so that it can persist across multiple runs of the program.

5. Tracking Progress:

- Calculate and display the total calories burned for the current week and compare it with the weekly goal.
- Use loops to sum up all the logged exercises for the current week and compare it with the goal.

6. Error Handling:

- Ensure the program handles invalid inputs (e.g., non-numeric input for calories, invalid date formats).

Bonus Features (Optional):

- Exercise Recommendations: Based on the user's remaining calories to burn for the week, recommend exercises.
- Graphical Display: If familiar with libraries like matplotlib, you can graph the user's calorie burn progress.

Deliverables: A single zipped file comprising 2 files:

1. Python script (.py file).
2. A word document report containing:
 - Brief explanation of the project
 - A "How to Use" section
 - A table enumerating various data structure used in the program, and the reason for their selection over one other possible candidate.
 - Include screenshots showing:
 1. Logging exercises.
 2. Viewing logged exercises for a given day.
 3. Setting and displaying weekly goals.
 4. Tracking weekly progress.

Sample Workflow:

1. Main Menu Example:
 1. Log an Exercise
 2. View Logged Exercises
 3. Set Weekly Calorie Goal
 4. Track Weekly Progress
 5. Exit

Enter your choice:

2. Log an Exercise Example:

Input:

Enter exercise type (e.g., running): Running

Enter the duration in minutes: 30

Enter calories burned per minute: 10

Output:

Exercise logged successfully for [current date].

3. File Structure:

Example log file:

2024-10-05 | Running | 30 minutes | 300 calories

2024-10-05 | Cycling | 20 minutes | 200 calories

Evaluation Criteria:

- Does the report covers the requirements mentioned in the deliverable section? (40%)
- Correctness: Does the program perform the required tasks without errors? (30%)
- Reusability: Are functions used appropriately to avoid code repetition? (10%)
- Error Handling: Does the program handle incorrect inputs without crashing? (10%)
- Clarity: Is the code well-commented and the logic easy to follow? (10%)